

● MEDIUM

● GEOMETRY AND TRIGONOMETRY

Area and volume

03

10 questions · ~15 min

Q1

GEOMETRY AND TRIGONOMETRY

A cylinder has a diameter of 8 inches and a height of 12 inches. What is the volume, in cubic inches, of the cylinder?

- A. 16π
- B. 96π
- C. 192π
- D. 768π

Q2

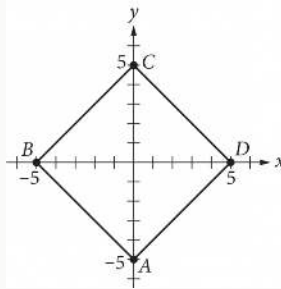
GEOMETRY AND TRIGONOMETRY

The length of one side of square M is 6 times the length of one side of square N. The area of square N is 49 square feet. What is the area, in square feet, of square M?

- A. 1,764
- B. 504
- C. 294
- D. 252

A right circular cylinder has a volume of 432 cubic centimeters. The area of the base of the cylinder is 24 square centimeters. What is the height, in centimeters, of the cylinder?

- A. 18
- B. 24
- C. 216
- D. 10,368



In the xy -plane shown, square ABCD has its diagonals on the x - and y -axes. What is the area, in square units, of the square?

- A. 20
- B. 25
- C. 50
- D. 100

A sphere has a radius of $\frac{17}{5}$ feet. What is the volume, in cubic feet, of the sphere?

- A. $\frac{5\pi}{17}$
- B. $\frac{68\pi}{15}$
- C. $\frac{32\pi}{5}$
- D. $\frac{19,652\pi}{375}$

The table gives the perimeters of similar triangles TUV and XYZ , where TU corresponds to XY . The length of TU is 18.

	Perimeter
Triangle TUV	37
Triangle XYZ	333

What is the length of XY ?

- A. 2
- B. 18
- C. 55
- D. 162

Q7

GEOMETRY AND TRIGONOMETRY

A cube has an edge length of 41 inches. What is the volume, in cubic inches, of the cube?

- A. 164
- B. 1,681
- C. 10,086
- D. 68,921

Q8

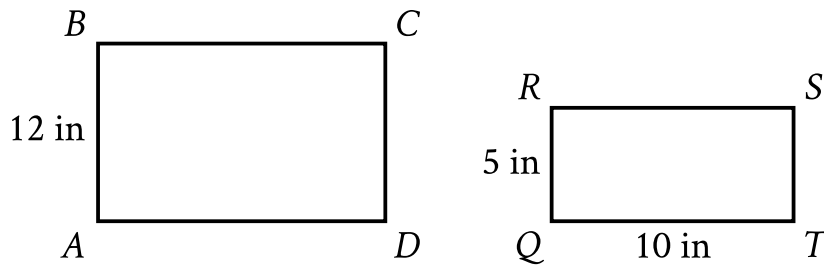
GEOMETRY AND TRIGONOMETRY

The length of the edge of the base of a right square prism is 6 units. The volume of the prism is 2,880 cubic units. What is the height, in units, of the prism?

- A. $4\sqrt{30}$
- B. 36
- C. $24\sqrt{5}$
- D. 80

A circle has a radius of 43 meters. What is the area, in square meters, of the circle?

- A. $\frac{43\pi}{2}$
- B. 43π
- C. 86π
- D. $1,849\pi$



Note: Figure not drawn to scale.

- Rectangle upper A upper B upper C upper D is labeled as follows:
 - The length of side upper A upper B is 12 inches.
- Rectangle upper Q upper R upper S upper T is labeled as follows:
 - The length of side upper Q upper R is 5 inches.
 - The length of side upper Q upper T is 10 inches.
- A note indicates the figure is not drawn to scale.

Rectangles $ABCD$ and $QRST$ shown are similar, where A , B , C , and D correspond to Q , R , S , and T , respectively. What is the length, in inches in, of $\overline{AD}$?

- A. 60
- B. 24
- C. 17
- D. 10